Zhaoxiang Feng

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Seeking Software Engineering (ML Systems, LLM Inference) and ML Research Internships | Available May–Aug. 2027

Education

The University of Texas at Austin — Austin, TX
Master of Science in Computer Science — *Aug. 2026 – Present*
Coursework: Post-Training of LLMs, ML for Systems, Natural Language Processing, Mathematics in Deep Learning

University of California San Diego — San Diego, CA
B.S. in Data Science; B.S. in Probability & Statistics; GPA: 3.84/4.00 — *Sep. 2022 – Mar. 2026*

Research Experience

Hao AI Lab, UC San Diego — San Diego, CA
Research Assistant, Advisor: Prof. Hao Zhang — *Sep. 2025 – Present*

- Led **JetSpec++** (first author, equal contribution; under review at ICLR 2027): **1,361 tokens/s** single-request on GB200, up to **7.45**× over autoregressive decoding and **1.54**× over DSpark, by co-designing self-refining parallel drafting, foresight drafting, and pipelined draft-verify overlap
- Built JetSpec's standalone **LLM inference engine** (**paged KV cache**, continuous batching, **Triton** tree attention, **CUDA-graph** verification): up to **1,150 tokens/s** single-stream on one B200
- Second author of **JetSpec** (under review at ICLR 2027), which drafts a full token tree in one forward pass: up to **9.64**× end-to-end speedup over autoregressive decoding (Qwen3-8B, MATH-500, H100)

Q-Lab, UC San Diego — San Diego, CA
Research Assistant, Advisor: Prof. Lianhui Qin — *Jan. 2025 – Jun. 2026*

- Contributed to **ArcMemo**'s typed, composable memory for reusable reasoning concepts and its reasoning-guided concept selection; **Runner-Up, ARC Prize 2025 Paper Awards**
- Built a **concept-to-puzzle data pipeline** that adapts the open-source BARC generator to turn hand-written ARC concepts into executable puzzle generators and validated helper puzzles

Wang Lab, UC San Diego — San Diego, CA
Student Researcher, Advisor: Prof. Wei Wang — *Jun. 2025 – Jun. 2026*

- Developing **TrustPPI**, a reliability signal from embedding-perturbation stability for protein interaction models; **0.66–0.83 AUROC** error detection across three architectures; manuscript in preparation

Projects

DFlash on Google Cloud TPU | *JAX, vLLM tpu-inference, TPU v4/v5p/v6e* — Code | Google Developers Blog

- Ported DFlash block-diffusion speculative decoding from GPU to TPU in JAX with the Google Cloud TPU team, from a standalone draft-verify loop with dual KV cache to vLLM serving integration
- Upstreamed a **2.8K-line PR** into vLLM tpu-inference (#1868, merged): DFlash draft model, concat-attention layer, JAX and torchax proposers, and unit tests
- Reached **3.13× average speedup** over nine benchmarks on TPU v5p (Qwen3-4B, up to 5.72× on MATH-500), and **2.29**× end-to-end vLLM serving speedup vs. 1.30× for EAGLE-3 (Llama-3.1-8B)
- Showed that TPU verification cost stays flat in draft block size through **K=1024**, which makes much wider draft blocks nearly free to verify

LLM Pre-training System | *PyTorch, DDP, ZeRO-1, FlashAttention-2* — Sep. 2025 – Dec. 2025

- Built the training stack for a **1.8B-parameter** model on **8 NVIDIA B200 GPUs**, using BF16 and torch.compile; pre-trained on **117B tokens in 55 hours** (about **630K tokens/s** steady-state, **36% MFU**)
- Implemented **SFT and DPO** post-training on Alpaca and HH-RLHF

Technical Skills

Languages: Python, SQL, Java, R, JavaScript
ML Systems: PyTorch, JAX, Triton, vLLM, Hugging Face Transformers, DDP, ZeRO-1, CUDA graphs
Hardware & Tools: NVIDIA B200/GB200/H100, Google Cloud TPU v4/v5p/v6e, Git, Linux, Docker